

# Failure-Guided Sim2Real2Sim on a Low-Cost Manipulator

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## I. PROBLEM STATEMENT

Policies trained in simulation can fail in ways that the simulator never exposes. Uniform domain randomization spends much of its sampling budget on scenarios the policy already handles, while configurations that cause failures on hardware remain rare. Sim2Real2Sim addresses this gap by feeding real-world outcomes back into the simulator to refine the training distribution. Existing approaches often assume that real-world trials are inexpensive and demonstrate their methods on research-grade platforms equipped with force sensing. We instead ask whether the Sim2Real2Sim loop remains effective in the opposite regime: a hobby-grade arm with substantial backlash, no force feedback, and only about 150 real-world trials available over a semester. The training-distribution update is fully automatic, while scene resets remain manual.

## II. APPROACH

This is an **application-oriented** project. We apply the **Sim2Real2Sim** strategy introduced by Chang and Padir [1] to a new robotic platform and data regime, following the closely related **Fail2Progress** approach [2], which records real-world failure states and generates corresponding simulation environments for retraining.

### A. Task

**Planar pushing:** The SO-101 pushes one of six rigid objects from a randomized initial pose to a target pose. This task keeps the arm’s unreliable gripper out of the critical path while providing a continuous outcome measure rather than a binary success/failure signal.

### B. Loop

We train a policy in MuJoCo under uniform domain randomization, deploy it to the physical SO-101, and log each trial’s scene parameters and outcome. We then fit a density model to the failing configurations and use it to resample the next training round from a fixed mixture of the uniform sampler and the failure density. We repeat this process for three cycles.

### C. What we adapt

Only observable scene factors are adapted: object start pose, goal pose, object identity, approach direction, and clutter. Dynamics parameters, including friction, joint damping, and actuator delay, remain uniformly randomized throughout, since real-world failures cannot be reliably attributed to these factors without force sensing.

## III. SIMULATION ENVIRONMENT

We use MuJoCo with an SO-101 model converted from URDF to MJCF and subsequently tuned with corrected inertial parameters and decomposed convex collision meshes. We chose MuJoCo over Isaac Lab because our policy is state-based rather than vision-based, eliminating the need for high-fidelity rendering. MuJoCo also runs efficiently on CPU, allowing team members without GPU access to develop locally. If simulation throughput becomes a bottleneck, we can leverage MJX for accelerated sampling.

## IV. EVALUATION

The primary metric is final object pose error, in position and orientation, measured on held-out start and goal configurations that never enter the failure log. We use continuous error rather than binary success because at 60 trials per condition a binary metric can only resolve differences of roughly 15–20 percentage points.

We compare three conditions at matched simulation compute:

- 1) Uniform domain randomization (baseline).
- 2) Adaptation guided by failures observed *in simulation*.
- 3) Adaptation guided by failures observed *on hardware*.

Condition B is essential. Without it, a win for C over A could be explained by training on a harder distribution rather than by real-world evidence; the B-versus-C comparison isolates the value of the real data. We budget 60 evaluation trials per condition plus roughly 150 across the three collection cycles, evaluate conditions in randomized interleaved blocks within sessions so that method is not confounded with hardware wear, and report pose error against real trials consumed with confidence intervals.

## REFERENCES

- [1] P. Chang and T. Padir, “Sim2real2sim: Bridging the gap between simulation and real-world in flexible object manipulation,” 2020. [Online]. Available: <https://arxiv.org/abs/2002.02538>
- [2] Y. Chebotar, A. Handa, V. Makoviychuk, M. Macklin, J. Issac, N. Ratliff, and D. Fox, “Closing the sim-to-real loop: Adapting simulation randomization with real world experience,” 2019. [Online]. Available: <https://arxiv.org/abs/1810.05687>

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